Test time – 1 hour

Each question 10 marks

1) a) A sequence of numbers F(n) is defined such that each number in the sequence is the sum of two preceding numbers in the sequence. Also F(0) = 0 and F(1) = 1. Could you write pseudo code to create an array of size n containing the first n numbers in the sequence? (5 marks)

   b) Complete the following pseudo code to find square root of a number n: (5 marks)

```
{
int n;
double c1,c2,c3;

c1=0; c2=n;

while ( (c2-c1)>=0.000001))

{

c3= (c1+c2)/2;




}

cout<<" Square root of the number is"<<c3;

}
```

2) A stick of length 1 is randomly broken into two

   a. What is the average length of the smaller piece
   b. What is the average ratio of smaller piece to larger piece

   Use ln(2) = 0.693

3) Mr. A was born in the month of March. What is the expected number of people he needs to meet
   a. To find some one who is born in March
   b. To have a 50-50 chance of finding someone born in March

   Assume probability of being born in any month is the same = 1/12
   Assume $ln(1+x) = x$ or $ln(1+x) = x - x^2/2$ based on approximation required for small x
   Also assume ln(2) = 0.693

4) Evaluate

$$\int \int (3 - x - y)\ dydx$$

   over the area x>=y, 0<=x<=1, 0<=y<=1. Show the steps of the evaluation.

5) Suppose X and Y are two random variables having a joint pdf as below

$$f(x, y) = \beta\, x^2 y \quad ; \quad 0 \le x \le 1\, ; 0 \le y \le 1$$

   $\beta$ is a constant. What should be the value of $\beta$ so that X and Y are independent. Also, can $\beta$ take any value such that X and Y are dependent?

   Provide the steps of the solution.

6) Find $\int_{-\infty}^{\infty} x^2 e^{-x^2}\, dx$. Hint 1 - look at $d(x\, e^{-x^2})$, Hint 2 – remember the pdf of standard normal variable will integrate to 1 over the range (-∞, ∞)

7) Let C be a function of K and $\sigma$, where $\sigma$ is in turn a function of K. In other words, $C = C(K,\ \sigma(K))$. Using 2-dimensional Taylor series expansion prove

$$\frac{dC}{dK} = \lim_{\Delta \to 0} \frac{C(\ K+\Delta,\ \ \sigma(K+\Delta)) - C(K, \sigma(K))}{\Delta} = \frac{\partial C}{\partial K} + \frac{\partial C}{\partial \sigma}\frac{\partial \sigma}{\partial K}$$

8) Shares of 3 companies are being sold in the market. Let us denote values of these at a given time t as $S_1(t)$,$S_2(t)$,$S_3(t)$. You own 1 share of the first company $S_1(t)$. Mathematically the share price follows the process

$$S_1(t) = 0.5\ Z_1(t) + 0.9\ Z_2(t)$$

   Here $Z_1(t)$,$Z_2(t)$ are independent standard normal N(0,1) random variables. Your investment is risky (value is not a predictable constant) as z1 and z2 are random variables. To remove riskiness, you plan to buy other two stocks $S_2(t)$, $S_3(t)$ in quantities $\alpha$ ,$\beta$ respectively. What should be the values of $\alpha$ and ? The processes for $S_2(t)$, $S_3(t)$ are

$$S_2(t) = 0.35\, Z_1(t) - 1\, Z_2(t)$$
$$S_3(t) = -2.05\, Z_1(t) + 1.2\, Z_2(t)$$

9) You have points A(0,a) B(1,b). How should you solve for point C such that C(c,0) and AC+CB is minimized?

10) A virus affects 0.1% of people. A diagnostic test to detect the virus has been designed. The diagnostic test performs such that anyone having the virus will test positive. However, 5% of the people will be declared positive even if they do not have the virus. A randomly selected individual tests positive. What is the probability person actually has the virus?